

Topic 16: The Laplace Transform (The Time-to-Frequency Machine)

Professor Leonard Master Series · Laplace Transforms

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: The Frequency Machine

"Differential equations are hard because derivatives tie things together. What if there was a machine that could turn derivatives into simple multiplication by s ?"

That machine is the **Laplace Transform**:

$$\mathcal{L}f(t) = F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

Under this transform:

- $\mathcal{L}y' = sY(s) - y(0)$
- $\mathcal{L}y'' = s^2Y(s) - sy(0) - y'(0)$

Notice what happened: the calculus operation of differentiation turned into **multiplication by s** , and the initial values $y(0), y'(0)$ were automatically injected right into the algebra!

2. The 3-Step Solve-and-Invert Protocol

1. **Transform:** Take $\mathcal{L}$ of every term in the ODE.
2. **Algebra:** Group terms and solve for $Y(s)$:

$$Y(s) = \frac{\text{Numerator}(s)}{\text{Denominator}(s)}$$

3. **Invert:** Decompose using partial fractions or complete the square, then apply $\mathcal{L}^{-1}$ to return to $y(t)$.